Wave Basics

SYNOPSIS

A wave in physics is an umbrella term encompassing all things repetitive. When physicists hear the word “wave,” they think of oscillations, repeating patterns, cyclical behaviors, etc., that extend or spread over space and time. They also think of sinusoidal math functions (sines and cosines) that can describe all such repetitions. The sinusoidal functions require only three parameters to define them: Amplitude, wavelength, and frequency. You have learned about these parameters in waves and how they change specific wave features. In this note, we will review/go over these terms and summarize them.

Next, I want you to think about the energy in waves: If you see massive ocean waves crashing into shores, toppling over buildings, trees, and other structures, clearly, there is an enormous amount of energy in these waves! This energy originated from somewhere; perhaps there was an underground earthquake that generated tsunamis, so the energy in these ocean waves comes from the potential energy in rocked-up tectonic plates, which were under tremendous tension but abruptly released. This earthquake could have happened thousands of miles from the shores, but it could still inflict massive damage because the waves carry the energy! So, the water that does the waving is the medium that allows energy transmission. An ocean wave is an example of a mechanical wave that can carry energy as long as there is an expanse of water that can wave. Many wave phenomena are mechanical waves that require a medium that can wave. Waves in a string need the string as the medium to carry the energy of the shaker, as you have seen from the PhET simulation.

Now that you see that waves carry energy, what does the mechanical motion of the medium look like? In other words, what motion carries the energy? Let’s consider ocean waves again, with water as the medium that propagates the waves. If you can “chunk” sections of the ocean, the wave is nothing more than these chunks of ocean waters simply moving up and down individually, but there isn’t a “flow” of the ocean water from the earthquake to the shores. A perfect analogy is the “Mexican wave” you sometimes see in a sports stadium when spectators are perhaps bored. The key feature of the Mexican wave is that the waves sweep through and go around the whole stadium without halting. Note that crucially, the spectators are only standing up and sitting back down and do not shift from their seats! They only need to stand up and sit back down at the right moment, which dictates the wavelength and frequency of the Mexican wave. So, the waves in the medium transport the energy but not the material supporting the waves.

Ocean waves, sound waves, and waves in a string are all examples of mechanical waves where mechanical motions of the medium transmit the energy of the wave generator. Still, the materials the medium consists of are not transported. The ocean waves and waves in a string are just like the Mexican wave: The wave moves, let’s say, from left to right, but the spectators themselves move up and down. Here, the direction of the wave movement (“left to right”) is, thus, perpendicular to the medium’s mechanical motion (“up and down”). This type of wave is called **transverse wave** (trans = opposite).

The sound wave, in contrast, is a repeating pattern of high and low air pressure that moves through space. The pattern of high and low air pressure is the bunching up of air molecules (high pressure) and spacing out of air molecules (low pressure), which are caused by a chunk of air repeatedly moving “left – right, left – right” while the pressure wave (the repeating pattern of high and low pressure) goes from “left to right.” In other words, the direction of the wave movement (“left to right”) is parallel to the mechanical motion (left – right, left – right) of the medium (the chunk of air). This type of wave is called a **longitudinal wave** (longitudinal = along).

The last thing I want to discuss is the phase of the wave. When you hear the word “phase” in physics, envision “moon phase.” A phase of the moon is a specific part/place in the cycle of moon shapes. For example, the full moon phase is the spot in the cycle when the moon is a complete circular disc. A phase of a wave is a specific place in the wave, and that is it. You can say peak phase or trough phase, for instance. You can describe wavelength using phase: The distance between phases identical in an adjacent cycle. For period: The duration of phases identical in an adjacent cycle. For example, the wavelength is the distance between peak phases in an adjacent cycle.

Phase is useful when you are comparing two waves. Imagine two waves with the same wavelength (therefore the same frequency, why?), and you place one wave on top of the other. You find that all phases align; that is, the peak phase of one lies on top of the peak phase of the other. This is called **in-phase**. In contrast, you find, instead, the peak phase lines up with the trough phase. In other words, one wave is “shifted in phase” with respect to the other. When the peak overlays with the trough, this is **180-degree out-of-phase** or completely out of phase. When phases between two waves don’t align, we say waves are out of phase.

In this note, summarize important physics terms in waves using diagrams and equations. Answer the questions that follow.

BEFORE YOU READ ON — PREDICT

A cork floats a hundred metres off the beach and a swell is rolling in. By the time ten waves have gone past, **where is the cork?** Closer to shore, further out, or in the same place? Commit to an answer in your guided notes before the synopsis talks you into one. **Note 3** asks you the same question with the answer choices written out.

PERFORMANCE EXPECTATIONS

Explain amplitude, wavelength, and frequency in waves. Describe the velocity of a wave using equations in terms of wavelength and frequency. Describe mechanical waves and how mechanical waves carry energy (but not materials). Describe, compare, and contrast transverse and longitudinal waves and give examples of each. Use the phase of a wave to describe wavelength and frequency (period).

Instructions and Questions

1. Figs 1 and 2 represent a moving wave (the same wave; one is for spatial variation at a fixed time and one for temporal variation at a fixed position). Fig. 1 shows how the amplitude of the wave varies in space. Fig. 2 shows how the same wave’s amplitude varies with time.

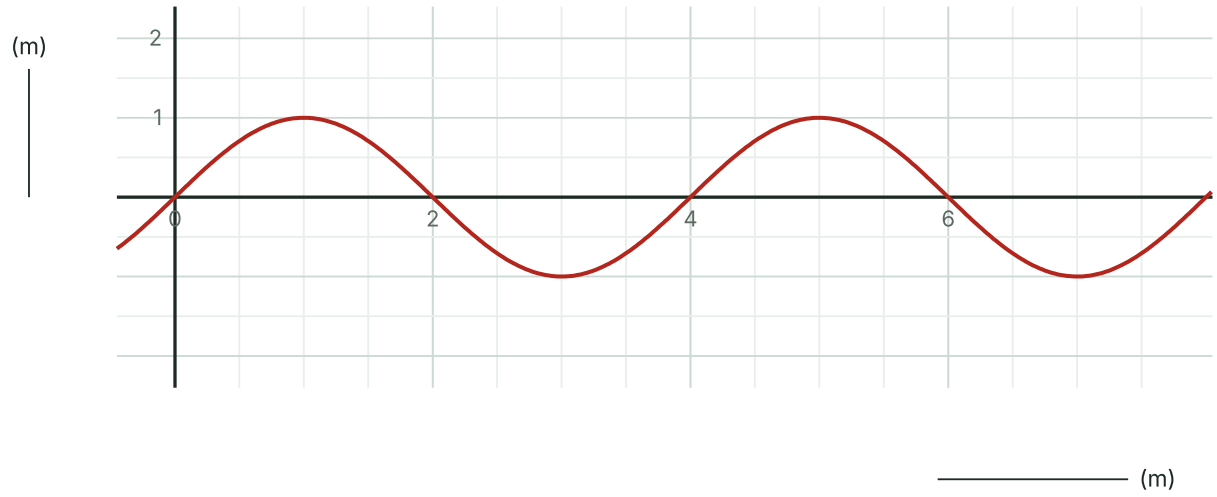


Fig 1. Amplitude vs. distance. A snapshot: the whole string at one instant.

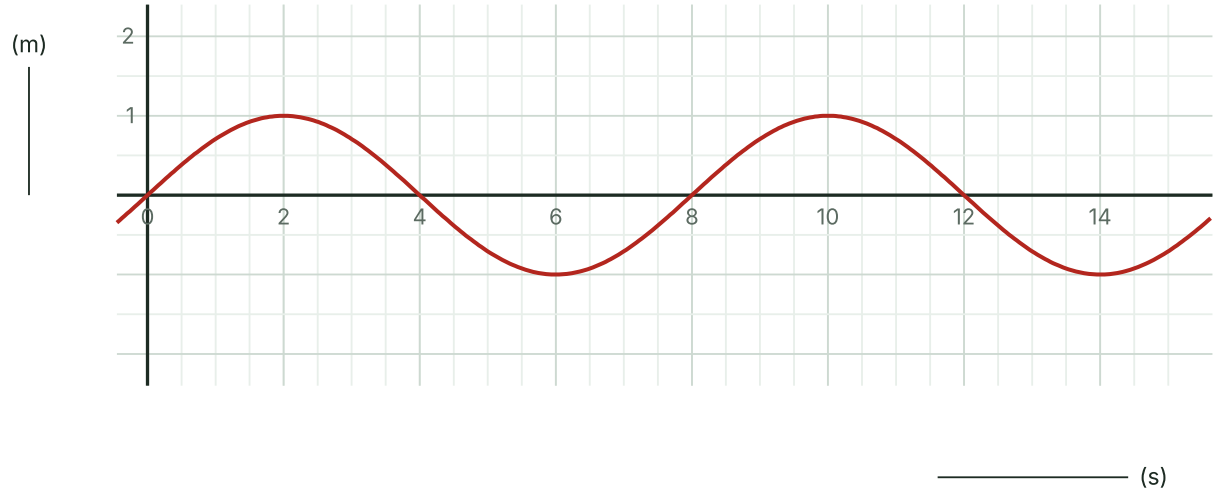


Fig 2. Amplitude vs. time. One point of the string, watched as time passes.

2. Label the axes in Fig. 1 and 2.

3. Read the amplitude from the figure.

Amplitude, A = _____ m

4. Indicate the amplitude (do it for both) on the figures.

5. Determine the wavelength (Fig. 1) and indicate in the figure.

Wavelength, λ = _____ m

6. Determine the period (Fig. 2) and indicate in the figure.

   Period, T = _____ s

7. What is the frequency of this wave? Also, show the unit.

   Frequency, f = _____

8. Imagine standing still (i.e., in a fixed position) on a beach and watching the waves coming at you, as the ocean waves lapping at your feet. How would you determine the speed at which these waves are arriving at you? Describe the method.

   Determine the velocity of a wave represented in Figs 1 and 2. State the equation first.

   _____

   _____

   _____

   _____

   v = _____ (Equation)

   v = _____ (show the unit)

9. What is a transverse wave? Draw and give examples.

   _____

   _____

10. What is a longitudinal wave? Draw and give examples.

11. What are the similarities and differences between transverse and longitudinal waves?

Similarities

– _____

– _____

– _____

– _____

Differences

– _____

– _____

– _____

– _____

– _____

12. What is a phase in waves?

13. Use the phase to describe the velocity of a wave. Complete the sentence. Or come up with your own phrase containing or using “phase.”

“The speed of a wave is the speed of a c_____ (8-letter word) phase.”

The 8-letter word: _____

Looking ahead

You now have the vocabulary the rest of the module runs on: amplitude, wavelength, period, frequency, the difference between a snapshot and a diary, transverse against longitudinal, and phase. Two of those carry a long way. **Note 3** is seventeen conceptual questions on everything these two notes have defined, and it is the right place to find out whether the vocabulary has actually stuck. After that, **phase** comes back in Note 8, where whether two waves reinforce or cancel is entirely a question of their phases; and **the medium sets the speed** becomes, in Note 7, an equation built from the tension and the mass of the string.

Where this sits. Notes 1 and 2 are the measuring and naming half of the module, and they are done. From Note 4 on, the same quantities are derived rather than read off a ruler: Note 4 asks what force makes one bead oscillate at all, and the sines and cosines of the synopsis come out of the answer.

BACK TO YOUR PREDICTION

The cork is in the same place. It has gone up and down ten times and travelled nowhere — the swell carried energy past it, not water. If you said “closer to shore,” you are in good company and you are thinking of a different effect: real ocean waves do nudge things shoreward a little, and that drift is not the wave. Question 10 of **Note 3** is this question with the answer choices written out, and it asks only about the up-and-down.